

TENGYUE ZHANG

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🎓 EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China 2023 – Present

B.S. in Physics (Major) & Computer Science (Minor) (Expected Jun. 2027)

- Member of the **Zhiyuan Honor Program** (Physics Track, Top 10% of Cohort)
- **Overall GPA:** 3.99 / 4.30 (Rank: 2 / 29, as of Sept. 2025)

♥ RESEARCH INTERESTS

AI for Science: Integrating AI with physical principles to enhance the performance of neural networks or solve physics problems. Key focuses include Quantum Machine Learning (QML) for variational quantum circuits and Physics-Informed Neural Networks (PINNs) for solving PDEs.

Multimodal Generation: Developing advanced generative models (e.g., Discrete Diffusion) for multimodal synthesis, with a specific focus on Text-to-Speech (TTS) currently.

🔬 RESEARCH EXPERIENCE

X-LANCE Lab, SJTU Jun. 2025 – Present

Research Assistant Advisor: Assoc. Prof. Xie Chen

Developing DiTAR-like TTS model using LocDiT architecture with patchification techniques and discrete diffusion methods based on F5-TTS framework.

ReThinkLab, SJTU Jul. 2025 – Oct. 2025

Research Assistant Advisor: Prof. Junchi Yan

Implemented Variational Quantum Algorithms (VQA) and Physics-Informed Neural Networks (PINNs) for solving PDEs; extended QuanONet framework for quantum-classical hybrid computations.

Zhiyuan Scholar Program, SJTU Jan. 2025 – Jun. 2027 (Expected)

Research Assistant Advisor: Prof. Hao Zheng

Investigating the spectroscopy of Higher-Order Topological Matter (HOTIs and HOTSCs) on STM platform.

⚙️ SKILLS

- **Programming Languages:** Python(Pytorch) > C++ > Julia = MATLAB.
- **Dev Tools:** Git/GitHub, L^AT_EX, Docker, Ubuntu / HPC user.
- **Machine Learning:** Self-studied Machine Learning, Deep Learning, Natural Language Processing. Now studying DRL(UCB CS294) and post-training techniques.
- **Generative Models:** Familiar with denoising diffusion models such as DDPM, Flow Matching and Masked Diffusion Model (MDM); Discrete tokenization methods such as VQ-VAE, FSQ, RVQ, etc.
- **Mathematics & Physics Basis:** Achieved A(+) across all physics-related courses (until Sept. 2025).

🏆 HONORS AND AWARDS

Meritorious Winner, Interdisciplinary Contest in Modeling (ICM) May 2025

Huatai Securities Technology Scholarship, Shanghai Jiao Tong University 2024